

# Swapnaj Khobragade

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## EDUCATION

**University of Mumbai**  
*B.E. in Computer Engineering*

**Expected 2028**  
*Mumbai, India*

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Bash  
**Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication  
**Cloud & DevOps:** AWS (EC2, S3, IAM, Lambda), Docker, Docker Compose, Terraform, Proxmox, k3s, Tailscale  
**Databases:** PostgreSQL, Redis  
**Monitoring:** Prometheus, Grafana, Loki, Grafana Alloy

## EXPERIENCE

**E-Cell** **Oct 2025 – Present**  
*Technical Member*

- Developed and maintained the official Entrepreneurship Cell website, improving maintainability and the club's digital presence.
- Supported technical operations for a national-level hackathon, including website updates, participant onboarding, and event coordination.

## PROJECTS

**HomeLab** | *Proxmox, Terraform, k3s, Grafana, Prometheus, Loki* **2026**

- Designed and operated a three-node self-hosted infrastructure using Proxmox, Docker, Kubernetes (k3s), and Tailscale for secure remote access.
- Automated infrastructure provisioning with Terraform and deployed a complete observability stack using Prometheus, Grafana, Loki, and Grafana Alloy.
- Implemented centralized logging, health monitoring, Telegram alerting, backup workflows, and disaster recovery runbooks for production-like reliability.

**Cloud Sentinel** | *Next.js, TypeScript, Python, Redis, PostgreSQL, AWS* **2026**

- Built a distributed AWS security auditing platform with a JWT-secured API gateway, Redis-backed job queue, Python workers, and PostgreSQL persistence.
- Implemented scheduled scans, retry queues, dead-letter handling, CIS benchmark mapping, and automated risk scoring for cloud security assessments.
- Designed the platform for secure, fault-tolerant asynchronous security scans across multiple AWS accounts.

**Aliasly** | *React, Node.js, PostgreSQL, Redis, Docker* **2026**

- Developed a containerized URL shortener supporting phrase-only aliases, password-protected links, optional expiry, and anonymous ownership.
- Implemented Redis caching, rate limiting, Google Safe Browsing integration, Cloudflare Turnstile, and real-time alias availability validation.
- Designed secure request handling to prevent authentication bypass while maintaining a fast and responsive user experience.

**Receipt** | *Python* **2026**

- Built a dependency-free XLSX parser using Python's standard library to convert bank statements into Cashew-compatible imports.
- Implemented a two-stage transaction classifier combining learned user corrections with regex-based categorization for improved accuracy.
- Provided both CLI and lightweight web interfaces while remaining dependency-free.

## CERTIFICATIONS

- |                                       |                                 |
|---------------------------------------|---------------------------------|
| • AWS Cloud Quest: Cloud Practitioner | AWS Educate   May 2026          |
| • Introduction to Linux (LFS101)      | The Linux Foundation   Jul 2026 |
| • Getting Started with Compute        | AWS Educate   Apr 2026          |